

Total No. of Questions : 5]

[Total No. of Printed Pages : 4

Roll No. _____

BS/Sem V/146

B. Sc. (Hons) Semester V Examination

2 0 1 7 – 1 8

PHYSICS

Paper No. BPT-501 : Mathematical Physics

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks

Answer all questions

Unless otherwise specified, the mathematical symbols have their usual meanings

1. Answer any six of the following :

6×3 = 18

- (a) Explain with examples, what is the difference among scalar matrix, diagonal matrix and identity matrix.
- (b) Find the singular points and the nature of the singularities for the equation :

$$x^2(x-2)^2 \frac{d^2y}{dx^2} + 2(x-2) \frac{dy}{dx} + (x+3)y = 0$$

- (c) Show that every second-order tensor can be expressed as the sum of a symmetric tensor and a skew-symmetric tensor.
- (d) Find the value of Legendre polynomial $P_n(x)$ for $n = 2$.
- (e) Define Dirac delta function $\delta(x)$ and show that $\delta(ax) = a^{-1}\delta(x)$; $a > 0$.
- (f) Define Wronskian $W[f_1(x), f_2(x)]$ of two differentiable functions $f_1(x)$ and $f_2(x)$. What is its significance?
- (g) Find the first non-zero term of Fourier expression of $f(x) = x^3$ for $-\pi < x < \pi$.

(Turn Over)

2. (a) Verify that following matrix is Hermitian:

$$H = \begin{pmatrix} 2 & 3-i \\ 3+i & -1 \end{pmatrix}$$

Find its eigenvalues and eigenvectors. Also write a Unitary matrix U that diagonalizes H by a similarity transformation. **9**

(b) Determine metric tensor $g_{\mu\nu}$ in spherical coordinates. **6**

Or

- (a) Show that $\varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$. Here the symbols have their usual meanings. **7**
- (b) Resolve the Cartesian unit vectors into spherical polar components and then express $x\hat{i} + 2y\hat{j} + yz\hat{k}$ in spherical polar coordinates. **8**

3. (a) If $y_1(x) = J_1(x)$ is the first solution of following differential equation :

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0$$

Find the second linearly independent solution using the integral method. **7**

- (b) Starting from the relation

$$\exp\left\{\frac{x}{2}\left(z - \frac{1}{z}\right)\right\} = \sum_{n=-\infty}^{n=+\infty} \{J_n(x)z^n\}$$

show that

$$J_n(x+y) = \sum_{k=-\infty}^{k=+\infty} \{J_k(x)J_{n-k}(y)\}$$

6

Or

- (a) If

$$\exp(A) = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

show that

$$\exp(A) = e^x \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}$$

where $A = \begin{pmatrix} x & x \\ x & x \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. **6**

- (b) If $\psi_n(x) = e^{-x^2/2}H_n(x)$, where $H_n(x)$ is Hermite's polynomial of degree n , then prove that—

$$\int_{-\infty}^{\infty} \psi_m(x)\psi'_n(x) dx = \begin{cases} 0 & , \text{ if } m \neq n \pm 1 \\ 2^{n-1}n!\sqrt{\pi} & , \text{ if } m = n - 1 \\ -2^n(n+1)!\sqrt{\pi} & , \text{ if } m = n + 1 \end{cases}$$

7

4. (a) A periodic wave function $f(x)$ of periodicity 2π vanishes in the region $-\pi < x < 0$ and it is represented by a straight line ($y = x$) in the region $0 < x < \pi$. Sketch a graph of $f(x)$ and find its Fourier series. **8**
- (b) A rigid body is rotating about a fixed axis with a constant angular velocity $\vec{\omega}$ that lies along z -axis. Express $\vec{r}$ in circular cylindrical coordinates and evaluate (i) linear velocity $\vec{v}$ and (ii) $\vec{\nabla} \times \vec{v}$. **5**

5. (a) Show that

$$\frac{1}{(1-t)} \exp\left(\frac{-xt}{1-t}\right)$$

is the generating function of Laguerre polynomials $L_n(x)$. **8**

(b) Find the Fourier series of following function :

$$f(x) = x \cos x, \quad -\pi < x < \pi$$

9.5

ExternalPortals&AcademicResources:

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>